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AMW-YOLOv8n: Road Scene Object Detection Based on an Improved YOLOv8

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Abstract: This study introduces an improved YOLOv8 model tailored for detecting objects in road scenes. To overcome the limitations of standard convolution operations in adapting to varying targets, we introduce Adaptive Kernel Convolution (AKconv). AKconv dynamically adjusts the convolution kernel's shape and size, enhancing the backbone network's feature extraction capabilities and improving feature representation across different scales. Additionally, we employ a Multi-Scale Dilated Attention (MSDA) mechanism to focus on key target features, further enhancing feature representation. To address the challenge posed by YOLOv8's large down sampling factor, which limits the learning of small target features in deeper feature maps, we add a small target detection layer. Finally, to improve model training efficiency, we introduce a regression loss function with a Wise-IoU dynamic non-monotonic focusing mechanism. With these enhancements, our improved YOLOv8 model excels in road scene object detection tasks, achieving a 5.6 percentage point improvement in average precision over the original YOLOv8n on real road datasets.

Keywords: road scene detection; YOLOv8; adaptive kernel convolution; multi-scale dilated attention; small target detection



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1. Introduction

In today's rapidly developing fields of autonomous driving and intelligent transportation, the importance of road scene object detection technology is increasingly prominent. It is not only the foundation for achieving vehicle autonomous navigation but also the key to ensuring driving safety. However, current object detection algorithms encounter significant challenges when handling small and multi-scale objects in complex road environments. Especially in urban road scenes where high-speed movement and visual occlusion are common, the accurate and rapid detection of various objects has become a major research focus.

One potential impact of accurately detecting road objects of varying sizes is that the smaller the detected object, the earlier the autonomous driving system can understand the traffic situation and make predictions or decisions. This will significantly enhance the safety of autonomous vehicles. For example, detecting small obstacles or pedestrians in advance can give the system more reaction time, avoiding potential collision risks. This not only improves the safety of autonomous vehicles but also increases the trust of passengers and other road users.

Previously, object recognition relied primarily on traditional machine learning algorithms, whose accuracy and performance were limited. Yet, as deep learning technology progresses and becomes widely adopted, the accuracy and performance of object recognition algorithms have significantly improved. Deep learning models have performed excellently in object recognition tasks and have been widely used in object recognition tasks in intelligent transportation systems.

Road object detection technology has undergone several development stages. Early object detection methods were mainly based on traditional computer vision algorithms,

such as Haar [1] feature classifiers, HOG [2] feature classifiers, and SIFT [3] feature classifiers. These methods required manual feature selection and tuning of feature extraction and classification algorithms, making it difficult to adapt to complex scenes, and their accuracy and robustness were relatively low.

In recent times, progress in deep learning has spurred the development of road object detection techniques utilizing this technology. These models are capable of autonomously learning features and excel in object detection and recognition tasks. Deep learning object detection frameworks are generally divided into region-based and single-stage detection [4]. Region-based algorithms detect objects by identifying regions within the image (region proposals), which are usually pre-selected, and then classified and localized by CNN. Region-based detection algorithms mainly include RCNN [5], Fast R-CNN [6], and Faster R-CNN [7]. Single-stage detection methods simultaneously identify and pinpoint objects within the image, usually faster than region-based algorithms but slightly less accurately. Single-stage detection algorithms mainly include SSD [8], RetinaNet [9], and the YOLO (You Only Look Once) [10] series. These methods are extensively applied in detecting objects in intricate road environments.

The YOLO algorithm employs a single-stage detection method, segmenting the image into grid cells and forecasting multiple bounding boxes and class probabilities per cell [11], enabling rapid traffic sign detection. Compared with traditional methods, the YOLO algorithm has higher detection speed and accuracy. In 2017, Huang et al. [12] introduced dense connection modules to convolutional neural network, enabling multiple reuses of each feature layer. In 2018, Redmon and Farhadi [13] proposed YOLOv3, which replaced YOLOv2's DarkNet_19 with DarkNet_53 and utilized a Feature Pyramid Network (FPN) [14] to generate multi-scale feature maps, enhancing small object detection. To minimize model parameters and computational loss, Bochkovskiy et al. [15] introduced YOLOv4 in 2020, featuring the CSPDarknet53 network and incorporating SPP (Spatial Pyramid Pooling) and feature fusion networks for compression and lightweighting. That same year, Ultralytics released YOLOv5 [16], which employs BottleneckCSP and Focus modules for feature extraction. In contrast to YOLOv4, YOLOv5 provides a more compact model, quicker processing, and reduced memory consumption. Additionally, it is the initial version built with the PyTorch framework, making it easy to use, train, and deploy as a lightweight model. The YOLOv7 [17] algorithm mainly focuses on optimizing the model architecture and training process, such as designing the Extended Efficient Layer Aggregation Network (E-ELAN) and using auxiliary heads during training, enabling it to run on low-end devices such as mobile or remote image processing devices. The YOLO series algorithms have now developed to the YOLOv8 version, which not only offers fast detection speed but also excels in detecting large- and medium-sized objects compared to mainstream two-stage detection algorithms.

To achieve high accuracy, high portability, and lightweight characteristics in road traffic scenes, this study introduces an enhanced YOLOv8 algorithm. By refining the feature extraction network, it can capture more detailed feature information and better handle the diverse and dynamic conditions of road objects. In addition, this study introduces Adaptive Kernel Convolution, which flexibly modifies the convolution kernel's shape and size; this allows the network to better adapt to various feature scales, thereby improving feature representation. To address the issue of small and difficult-to-recognize objects in complex traffic scenes, a small object detection layer is integrated into the network [18]; this layer extracts local features from larger feature maps, enhancing detail capture while preserving the accuracy of small object detection [19].

2. YOLOv8 Network Model

The YOLO series algorithms are renowned for their high accuracy and speed and are widely used in object detection tasks. YOLOv8 is the state-of-the-art (SOTA) algorithm in the YOLO series and surpasses other algorithms in both speed and accuracy [20]. YOLOv8 [21,22] is a versatile algorithm that combines pose estimation, object detection,

image classification, and instance segmentation. It incorporates many current SOTA technologies and is scalable, supporting other YOLO versions as well as more algorithms beyond YOLO.

YOLOv8 mainly consists of three parts: backbone, neck, and head [23]. **Backbone:** The backbone of YOLOv8 refers to the CSPDarkNet-53 network, particularly using the C2f (CSPLayer_2Conv) structure to replace the C3 module in YOLOv5. The C2f structure improves gradient flow [24], boosting model performance. In contrast to YOLOv5, the first convolutional layer's kernel in YOLOv8 has changed from 6×6 to 3×3 , which may affect the granularity of feature extraction. The backbone of YOLOv8 incorporates multiple convolutional layers, utilizing various kernel sizes and quantities to capture features across multiple scales. **Neck:** The neck of YOLOv8 employs a PAN-FPN structure, similar to YOLOv5, also referred to as a dual-path FPN. This design effectively merges feature information across various scales, enhancing detection accuracy. YOLOv8 incorporates the SPP module within the PAN-FPN to pool features at different scales, enabling the capture of targets of varying sizes. **Head:** YOLOv8's head section uses a decoupled head design, which splits the classification and detection heads [25], which helps the model focus better on their respective tasks and improve detection accuracy. Unlike YOLOv5, YOLOv8 has shifted from using anchors to an anchor-free method, eliminating the need for preset anchor boxes in object detection. **Loss Function:** YOLOv8 employs the Task-Aligned Assigner for matching positive and negative samples and incorporates Distribution Focal Loss (DFL) in its loss function. This approach enhances the prediction of bounding boxes and boosts detection accuracy.

3. Improvements to the YOLOv8 Detection Algorithm

The proposed AMW-YOLOv8n algorithm's network structure, illustrated in Figure 1, comprises four components: input, backbone, head, and output. This study enhances the backbone's feature extraction by introducing Adaptive Kernel Convolution (AKconv). AKconv dynamically adjusts the convolution kernel's shape and size, enabling the network to better handle features of different scales. This improvement enhances the model's capability to accurately detect targets of varying sizes, particularly in complex road scenes.

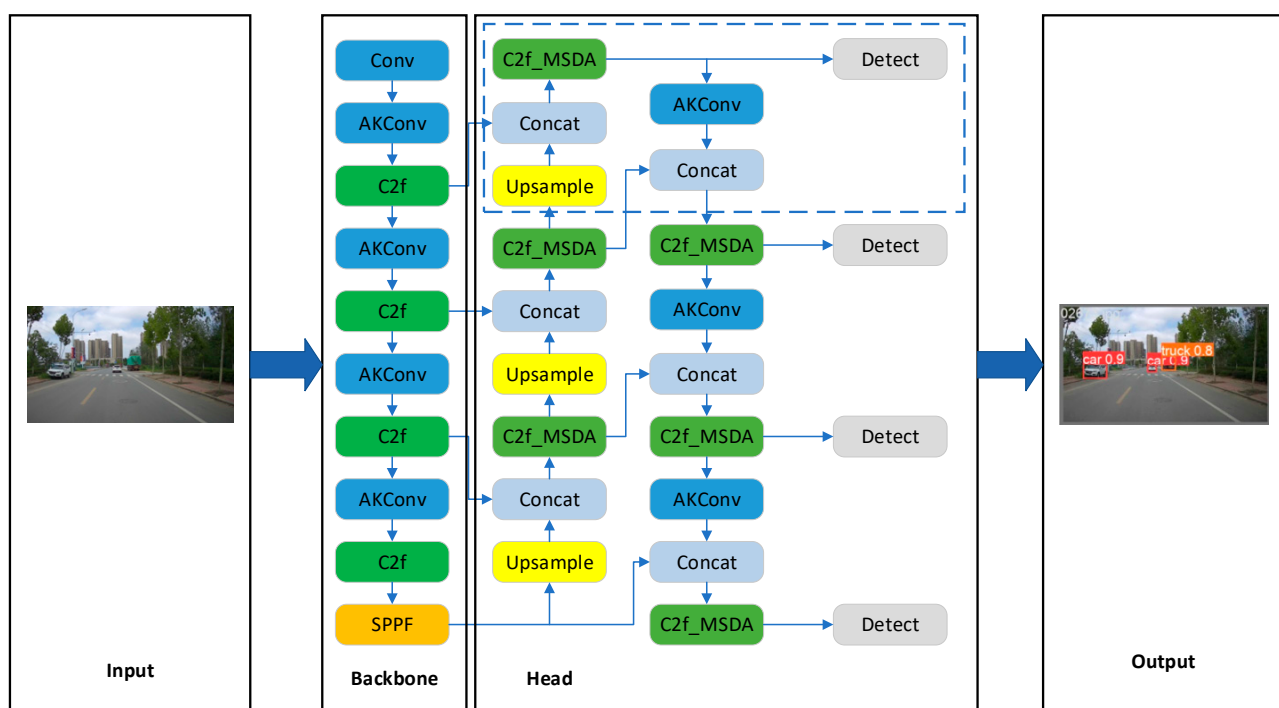


Figure 1. AMW-YOLOv8n network diagram.

To enhance the model's adaptability to targets of different scales, this study integrates a Multi-Scale Dilated Attention mechanism into the head network. This mechanism uses dilated convolutions to widen the receptive field without adding computational load, thus enhancing focusing on the key features of the target.

The head network utilizes a structure combining a Feature Pyramid Network and a Path Aggregation Network to enhance feature fusion capability [26]. This layer merges shallow and deep feature maps, boosting the model's capability to detect small targets and increasing accuracy in road scenes. Additionally, to boost training efficiency, the introduction of a regression loss function features a Wise-IoU dynamic non-monotonic focusing mechanism. This function dynamically adjusts the weight loss according to the size and shape of the target, speeding up model convergence and enhancing detection performance.

This chapter will sequentially discuss AKconv, Multi-Scale Dilated Attention, the small target detection layer, and the Wise-IoU loss function.

3.1. AKconv

In road scenes, targets such as vehicles and pedestrians are diverse and complex. Due to the variability in target size, shape, position, and orientation, traditional convolution operations face two main limitations: they are confined to local windows, preventing them from capturing information from other areas, and their sampling shapes remain constant. Additionally, the convolution kernel size is fixed at $k \times k$, resulting in a fixed square shape, with the number of parameters increasing quadratically as the size grows. Fixed sample shape square convolution kernels cannot adapt well to changing targets. These difficulties impair the model's accuracy in identifying targets, leading to incorrect or overlooked detections. To mitigate these issues, this study employs the AKconv network proposed by Xin et al. [27], reconstructing the convolutional network structure in YOLOv8 to boost detection capability.

AKconv mainly provides a flexible convolution mechanism that allows convolution kernels to have any number of parameters and sampling shapes. This method breaks through the limitations of traditional convolution; by moving beyond fixed windows and sampling shapes, convolution operations can more accurately adjust to various datasets and targets at different locations. The improvements of AKconv are as follows:

Flexible Convolution Kernel Design: AKconv enables convolution kernels to possess a flexible number of parameters, enabling them to adjust their size and shape according to actual needs, thereby more effectively adapting to target changes.

Initial Sampling Coordinate Algorithm: AKconv proposed a new algorithm to generate initial sampling coordinates for convolution kernels of different sizes, further enhancing its flexibility in handling targets of various sizes.

Adaptive Sampling Position Adjustment: To adapt to different target changes, AKconv adjusts the sampling positions of irregular convolution kernels through obtained offsets, improving the accuracy of feature extraction.

Reduction in Model Parameters and Computational Overhead: AKconv allows for the dynamic scaling of convolution parameters, optimizing performance for various hardware configurations, making it particularly suitable for lightweight model applications.

Figure 2 shows a detailed schematic diagram of the AKconv structure. The input image dimensions are (C, H, W), where C stands for the number of channels, and H and W correspond to the image's height and width, respectively. AKconv provides a distinct initial sampling shape for the convolution kernel. The convolution operation uses Conv2d to process the input image, while the offset component adjusts the initial sampling shape through learned offsets. This step is crucial for AKconv, allowing the convolution kernel shape to dynamically adjust to adapt to the image features. The resampling component adjusts the feature map according to the modified sampling shape. This adjusted feature map is then reshaped, convolved again, normalized, and finally activated using the SiLU function. The three rows below show the changes in sampling coordinates: The initial sam-

pling positions of the convolution kernel, indicated by the original coordinates, exclude any offsets. The offsets show the learned offsets that will be applied to the original coordinates, and the modified coordinates show the sampling coordinates after applying the offsets.

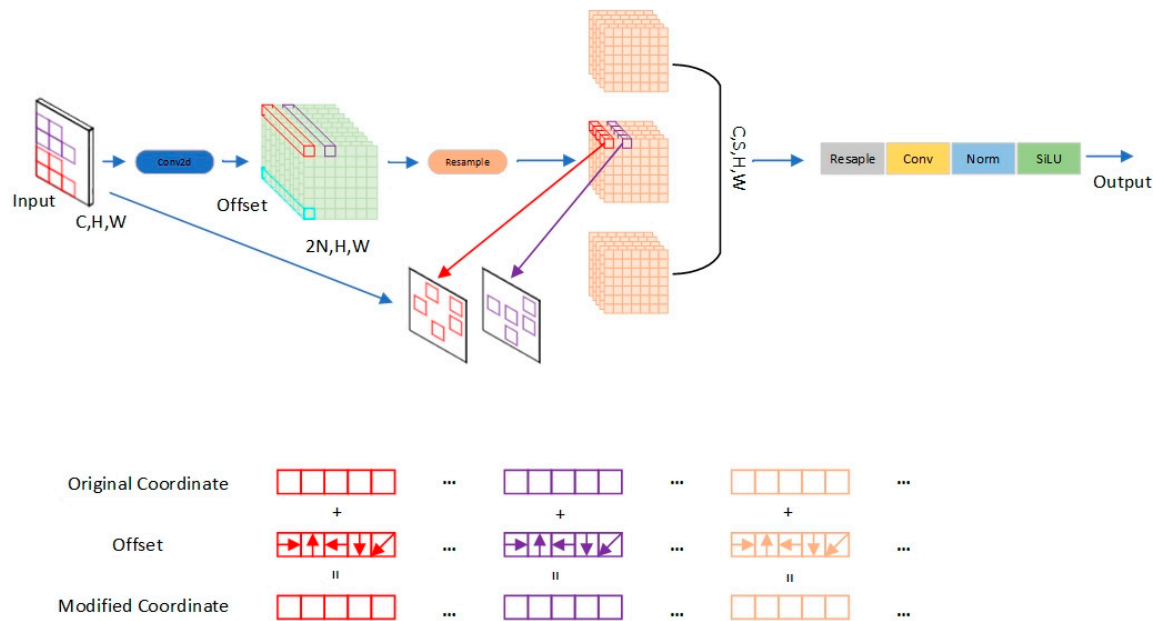


Figure 2. Schematic diagram of AKconv.

3.2. Multi-Scale Dilated Attention

MSDA, derived from Jiao et al.'s [28] transformer research, is an attention mechanism that models the interaction of local and sparse image patches within a small range. They found that in the shallow layers of Vision Transformers (ViTs), attention matrices are local and sparse, showing little correlation. This suggests significant redundancy in global attention modules. Building on this, they developed the DilateFormer model, which stacks MSDA blocks in the lower layers and adds global multi-head self-attention blocks in the upper layers. This design simulates long-range dependencies for high-level information processing while leveraging sparsity and locality for low-level tasks.

Figure 3 illustrates the MSDA structure. Initially, the channels of the feature map are divided into separate heads. Then, self-attention is applied to the colored patches surrounding the red query patch, with each head using different dilation rates. The concatenated features from various heads are then input into a linear layer. Typically, a 3×3 kernel is employed with dilation rates of 1, 2, and 3, leading to receptive fields of 3×3 , 5×5 , and 7×7 , respectively.

Specifically, the MSDA formulas are as shown in Equations (1) and (2):

$$h_i = \text{SWDA}(Q_i, K_i, V_i, r_i), \quad 1 \leq i \leq n, \quad (1)$$

$$X = \text{Linear}(\text{Concat}[h_1, \dots, h_n]), \quad (2)$$

Here, (r_i) represents the dilation rate of the (i) -th head, and (Q_i) , (K_i) , and (V_i) are the feature map slices input to the (i) -th head. The outputs for $(i = 1)$ are merged and directed to the linear layer for feature aggregation. MSDA employs varying dilation rates for each head to integrate semantic information across different scales within the receptive field. This approach minimizes redundancy in the self-attention mechanism without adding complexity or extra computational costs.

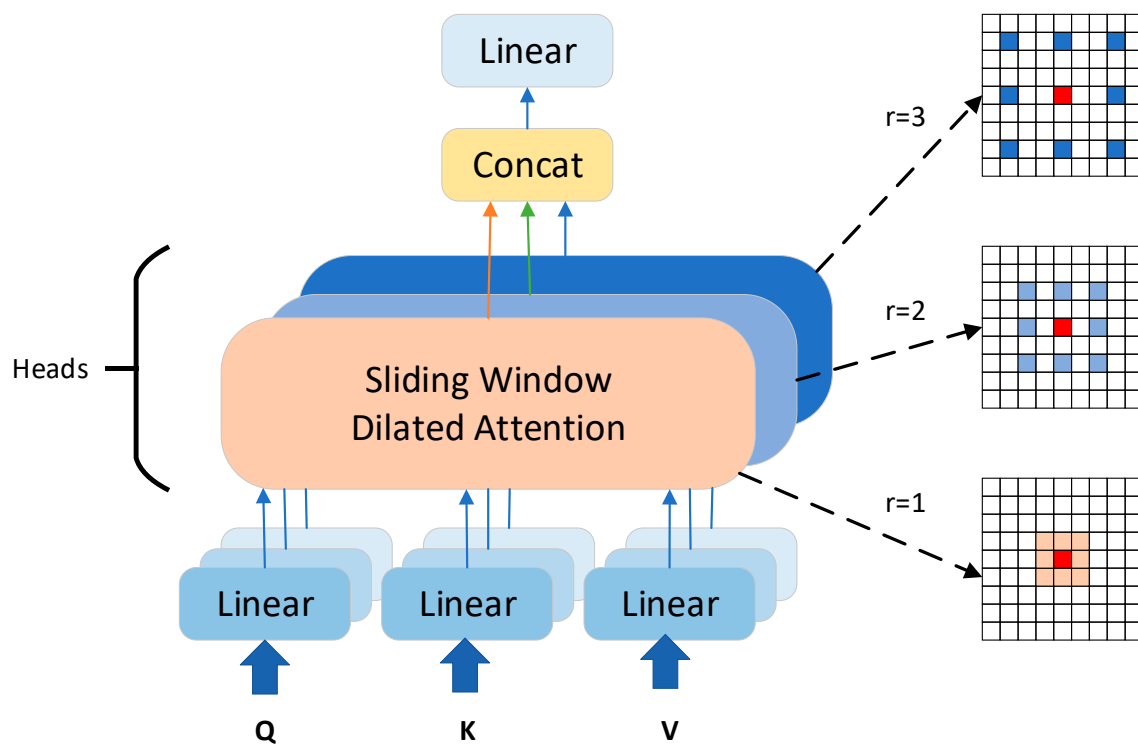


Figure 3. Diagram of Multi-Scale Dilated Attention convolution (MSDA) structure. The red square in the figure represents the query patch, while the surrounding colored patches are the representative patches selected in the Sliding Window Dilated Attention operation.

3.3. Small Target Detection Layer

In complex road environments, traditional object detection algorithms often overlook small objects such as pedestrians, small animals, and traffic lights. Ignoring these small objects can lead to serious safety risks. Therefore, designing a detection layer specifically for small targets is particularly important. YOLOv8's built-in head only uses three high-level feature maps from the backbone for feature fusion, which results in a large down sampling factor, making it challenging for deeper feature maps to capture small target features. Therefore, including a layer specifically for detecting small targets that concatenates shallow feature maps with deep feature maps for detection can enhance the network's focus on small target detection to boost overall performance [29]. This study presents a small target detection head to improve the model's performance in identifying small targets within dense scenes. This enhancement boosts the network's detection performance. The structure of this layer is illustrated in the dashed box in the head architecture shown in Figure 1.

The small target detection head incorporates the lower-level C2f feature map from the backbone network into the feature fusion process. By fusing high-dimensional low-level features with the original upper-level features, it can provide more detailed information, thereby enhancing the perception of small targets.

3.4. Optimization of the Loss Function

During the dataset creation process, overly large annotation boxes can lead to learning too much background information, while overly small annotation boxes can result in incomplete feature learning. Incorrect annotations can lead to learning incorrect features. It is inevitable that some low-quality examples will exist in the dataset, and reinforcing the regression of these low-quality bounding boxes can harm the model's detection performance [30]. YOLOv8's bounding box prediction function uses CIoU loss, which takes into account the aspect ratio of the predicted and actual boxes. However, CIoU loss does not resolve the issue of low-quality samples within the training dataset. To address this issue, this study uses Wise-IoU [31] to replace CIoU. Low-quality samples are inevitable in the

training dataset. Aggregation factors of anchor boxes, factors like distance and aspect ratio, can intensify the network's attention to these low-quality samples, resulting in a decline in generalization performance. Wise-IoU lessens the aggregation metric penalty when the anchor and target boxes significantly overlap. This minimizes intervention during pre-training, thereby enhancing the network's generalization ability. The formula for Wise-IoU is as follows:

$$F_{WIoU} = rR_t L_{IoU}, r = \frac{\beta}{\sigma \alpha^{\beta-\alpha}} \quad (3)$$

$$R_t = \exp\left(\frac{(x - x_{gt})^2 + (y - y_{gt})^2}{(\omega_g^2 + H_g^2)^*}\right) \quad (4)$$

$$\beta = \frac{L_{IoU}^*}{L_{IoU}} \in [0, +\infty) \quad (5)$$

Among them, $L_{IoU} = 1 - IoU \in [0, 1]$, which reduces R_t for high-quality anchor boxes, lessens the emphasis on the center distance point when there is significant overlap between the predicted and ground truth boxes. The outlier degree β describes the anchor box quality; a smaller β indicates higher quality. Assigning smaller gradient gains to anchor boxes with a large β can help mitigate the impact of low-quality samples. When $\beta = \sigma$, σ makes $r = 1$, and the anchor box receives the maximum gradient gain.

$\overline{L_{IoU}}$ is the moving average with momentum m . Since $\overline{L_{IoU}}$ is dynamic, the quality division standard of the anchor box is also dynamic, allowing the function to make the most suitable allocation strategy for the current gradient gain at any moment. At the beginning of training, initializing $\overline{L_{IoU}} = 1$ allows anchor boxes with $L_{IoU} = 1$ to have high gradient gains, setting the following momentum:

$$m = 1 - \sqrt[m]{0.05} \quad (6)$$

Using m to delay $\overline{L_{IoU}}$ from converging too quickly. In the later stages of training, Wise-IoU reduces gradient gains for low-quality anchor boxes, thereby prioritizing standard quality anchor boxes to enhance model performance [32].

4. Experimental Results and Analysis

This chapter assesses the proposed method's effectiveness using the AMW-YOLOv8n algorithm in road scene object detection tasks through experiments. This method involves training and testing the improved YOLOv8 alongside other mainstream object detection algorithms on the autonomous driving dataset, followed by a performance comparison of each algorithm. Additionally, ablation experiments are conducted on each improved algorithm module, including AKconv, MSDA, the small target detection layer, the Wise-IoU loss function, and the proposed YOLOv8 improved network with the aforementioned innovations. Images from different scenes in the dataset are also selected to compare the detection performance of different object detection methods' algorithms in actual scenarios. For each algorithm, the number of epochs is uniformly set to 100.

4.1. Experimental Environment and Parameter Settings

This experiment uses an NVIDIA A40 with 48 GB of memory. The operating system is a 64-bit Windows 10, version 22H2. PyTorch 1.12 is used as the deep learning framework, with CUDA version 11.3, cuDNN version 8.2, and Python version 3.8.19.

4.2. Dataset

The autonomous driving dataset used in this study is an internally collected dataset, with all images depicting real road scenes in China. This dataset includes scenes from multiple cities across China, captured under various weather conditions, at different times of the day, and in different locations. The training set contains 12,420 images, and the validation set contains 3295 images, with each image having a resolution of 1920×1080 pixels.

The dataset includes seven main categories of road scene objects: car, person, truck, bicycle, bus, traffic light, and motorcycle. Table 1 shows the number of objects of each type in the dataset and their minimum and maximum sizes in the images. Here, pic_num represents the number of images containing objects of that category; box_num represents the number of bounding boxes for objects of that category; Min Size represents the minimum size of objects of that category (in pixels); and Max Size represents the maximum size of objects of that category (in pixels).

Table 1. The number of objects of each type in the dataset used in this article and their minimum and maximum sizes in the image.

Models	Pic_Num	Box_Num	Min Size	Max Size
car	14,611	81,979	10.8 × 10.8	850.91 × 850.91
person	7789	19,996	9.23 × 9.23	768.36 × 768.36
truck	6652	10,276	14.41 × 14.41	873.55 × 873.55
bicycle	1550	2412	9.11 × 9.11	719.88 × 719.88
bus	2371	2897	20.2 × 20.2	895.78 × 895.78
traffic light	2836	7318	14.41 × 14.41	360.18 × 360.18
motorcycle	3340	5852	12.94 × 12.94	837.47 × 837.47

4.3. Evaluation Metrics

This study uses mean average precision (mAP), mAP50, mAP50-95, precision (P), and recall (R) as the evaluation metrics for the model. The values of these metrics are positively correlated with the model's performance. The calculation methods are as follows:

$$P = \frac{TP}{TP + FP} \quad (7)$$

$$R = \frac{TP}{TP + FN} \quad (8)$$

$$mAP = \frac{\sum AP}{N(class)} \quad (9)$$

TP represents true positives, *FP* denotes false positives, and *FN* indicates false negatives. *P* stands for precision, *R* for recall, and *AP* is the area under the *P*–*R* curve. *AP* is the sum of *AP* values for all categories, with *N(class)* representing the total number of categories. mAP50 is the mean *AP* for each category when the IoU between the predicted and actual boxes is set to 0.50. mAP50–95 represents the mean *AP* when IoU ranges from 0.50 to 0.95 with an interval of 0.05. *P* and *R* are both calculated at an IoU threshold of 0.50.

4.4. Comparative Experiments

In the comparative experiments of road scene object detection, this study uses seven classic object detection algorithms, including SSD, FCOS, Faster-RCNN, YOLOv5n, YOLOv7-Tiny, YOLOv8n, and the proposed AMW-YOLOv8n algorithm. To evaluate these algorithms, we use four metrics: Precision, Recall, mAP50, and mAP50–95. Table 2 presents the experimental results, clearly showing that the proposed improved YOLOv8 algorithm outperforms all others in every metric. Under the mAP50 metric, it is 4.8 percentage points higher than the second-best method, Faster-RCNN, and also outperforms other algorithms in precision and accuracy. The proposed algorithm is improved by 5.0 percentage points and 6.7 percentage points over YOLOv5n in the mAP50 and mAP50–95 metrics, respectively, and by 6.1 percentage points and 8.2 percentage points over YOLOv7-Tiny. Overall, the comparative experiments demonstrate that the proposed AMW-YOLOv8n algorithm significantly outperforms other algorithms in addressing object detection in complex road scenes.

Table 2. The experimental results of this algorithm are compared with those of other object detection algorithms.

Models	Precision/%	Recall/%	mAP50/%	mAP50–95/%
SSD	61.2	48.9	53.1	30.1
FCOS [33]	64.2	52	57.3	36.9
Faster-RCNN	66	56.8	58.7	40.5
YOLOv5n	65.6	52.8	58.5	35.6
YOLOv7-Tiny	63.5	53	57.4	34.1
YOLOv8n	64.5	52.5	57.9	37.8
AMW-YOLOv8n	66.8	57.5	63.5	42.3

4.5. Ablation Experiments

To assess the impact of the YOLOv8 model improvements, ablation experiments were performed on the autonomous driving dataset. These improvements include AKconv, MSDA mechanism, small target detection layer, and Wise-IoU loss function, demonstrating their effectiveness as well as necessity. “√” indicates that the method was used in the model, using the original YOLOv8 modules.

The ablation experiment results are shown in Table 3. The YOLOv8 model with the introduction of the MSDA mechanism improved mAP50 and mAP50-95 by 2.4 percentage points and 2.1 percentage points, respectively, compared to YOLOv8n. This indicates that the addition of MSDA effectively aggregates semantic information at different scales within the receptive field, improving feature extraction for targets of different scales and in blurry backgrounds. However, while the MSDA mechanism significantly improves model accuracy, it also increases computational complexity, leading to a decrease in FPS. The YOLOv8 model with modified AKconv also showed improvements in mAP compared to the original YOLOv8n. This suggests that AKconv improves feature extraction accuracy by adjusting the sampling positions of irregular convolution kernels based on obtained offsets, but it also increases computational complexity, resulting in a decrease in FPS. Adding the small target detection layer to the YOLOv8 model resulted in a 4 percentage point increase in mAP50. The significant improvement in accuracy also led to increased memory usage and a notable decrease in FPS. When only the loss function was improved, mAP50 increased by 0.6 percentage points, indicating that the optimized loss function converges faster and the prediction results are closer to the true values. In summary, for complex road scene object detection, each improvement enhances the detection accuracy of the original network to some extent, especially the small target detection layer and MSDA mechanism. Using all the improvements together maximizes the accuracy of the network model. Although the addition of attention mechanisms increases model complexity, computational load, and latency, it overall improves model accuracy. Moreover, an FPS of 228 is still sufficient for real-time detection in autonomous driving, enabling better detection of various targets in complex road scenes.

Table 3. Experiment results for each component.

AKconv	MSDA	Small Target Detection Layer	Wise-IoU	mAP50/%	mAP50–95/%	GFLOPs	FPS
				57.9	37.8	8.1	434
√				59.1	38	9.4	357
	√			60.3	39.9	9.1	344
		√		61.9	41.0	19.7	294
			√	58.5	37.7	8.1	403
√	√	√	√	63.5	42.3	12.6	228

4.6. Experimental Analysis

To illustrate the detection performance of the proposed algorithm, a comparative experiment was conducted. The four metrics of the proposed AMW-YOLOv8n algorithm, mAP50, mAP50–95, P , and R , are shown in Figure 4. The proposed model converges smoothly, and all four metrics, mAP50, mAP50–95, P , and R , show improvements compared to YOLOv8n, proving the effectiveness of the improvements in the algorithm.

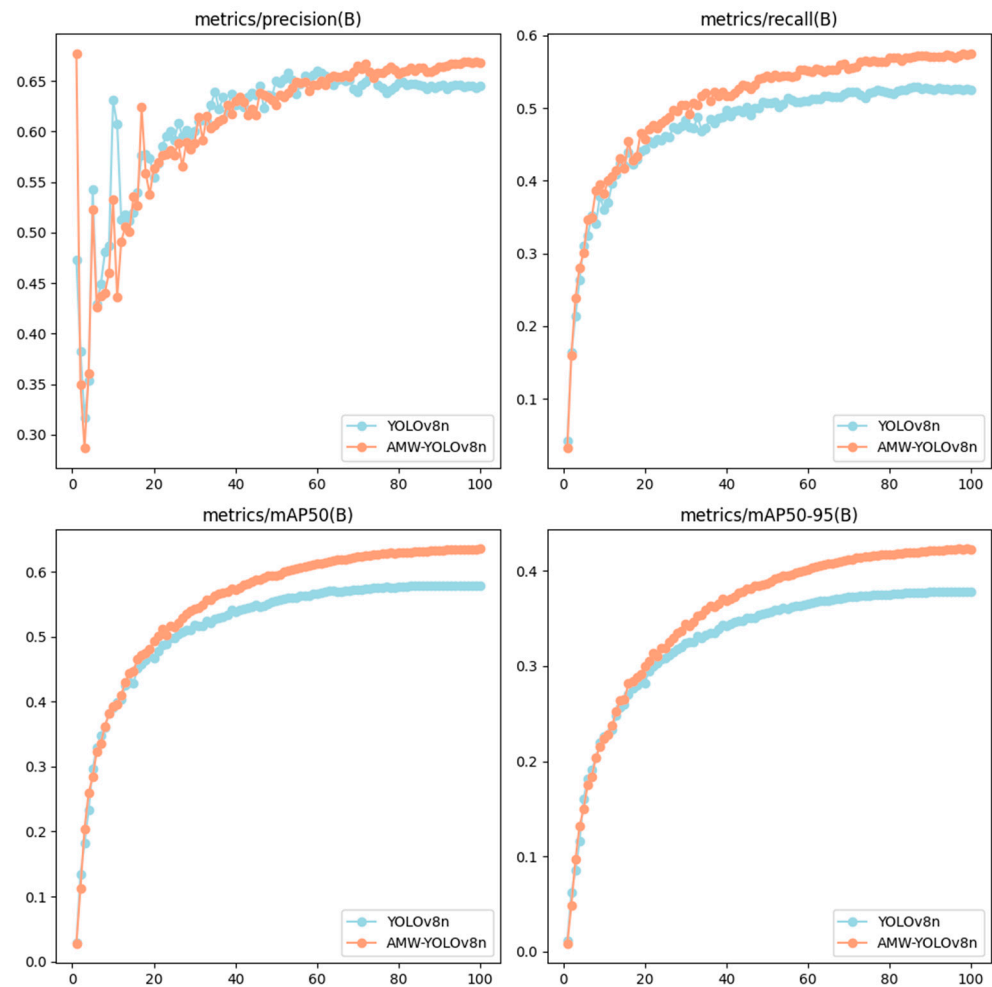


Figure 4. Comparison of training results between YOLOv8n and AMW-YOLOv8n.

Figure 5 compares detection effectiveness. Figure 5a displays the results of six images using the YOLOv8n algorithm, while Figure 5b shows the results using the proposed AMW-YOLOv8n algorithm. By comparing the detection results of six sets of the same scenes, it is evident that the AMW-YOLOv8n algorithm is capable of detecting a greater number of targets compared to other methods, including densely packed small targets and occluded people and vehicles. The proposed algorithm also enhances overall detection accuracy. The comparison of detection results demonstrates that the AMW-YOLOv8n algorithm excels in detecting targets in complex road scenes, effectively addressing occlusion and reducing missed detections of small targets.

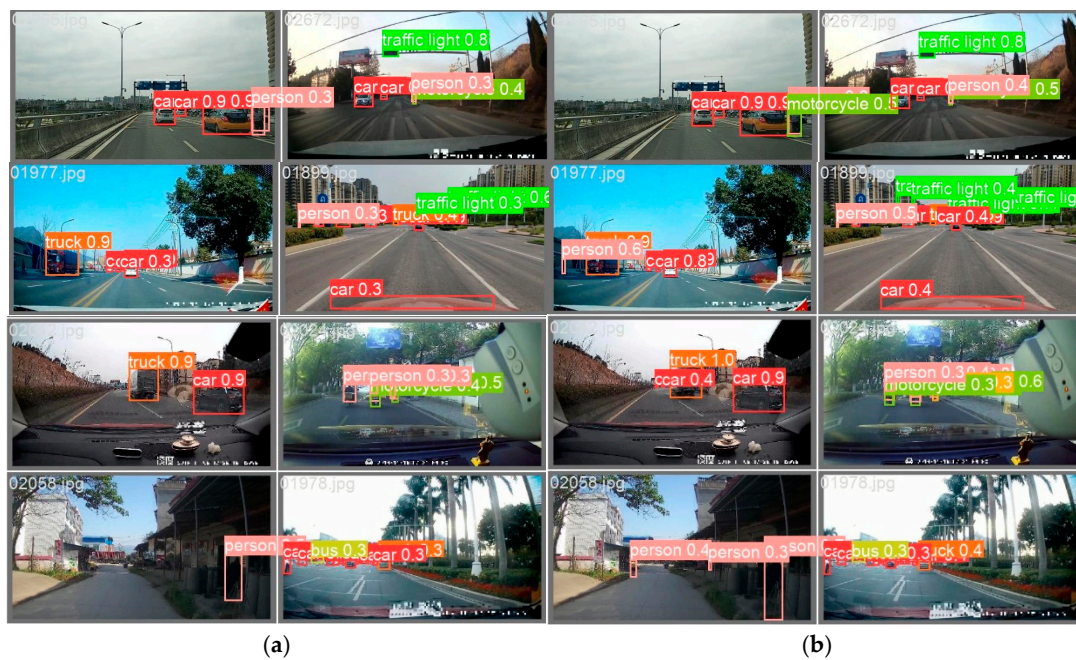


Figure 5. Detection effect comparison. (a) Detection results of YOLOv8n algorithm. (b) Detection results of AMW-YOLOv8n.

4.7. Data Analysis

To compare the detection performance of different improvements on various types of objects, this section records the mAP of seven categories of targets in the autonomous driving dataset, as shown in Figure 6. The experimental results show that after adding MSDA and the small object detection layer separately, the detection accuracy of small targets such as traffic lights significantly improved. After adding AKconv separately, the detection accuracy of all targets improved to varying degrees. To further demonstrate the detection performance of the proposed model for objects of different sizes, we classified and detected the objects in the dataset according to the classification standards of the COCO evaluation metrics, with the IoU for average precision set to 0.5. The detection results are presented in Table 4. Table 4 shows the average precision and average recall of YOLOv8n and AMW-YOLOv8n for large, medium, and small models. From Table 4, it can be seen that compared to YOLOv8n, AMW-YOLOv8n has improved the average precision and average recall for small targets by 10 percentage points and 12.7 percentage points, respectively. There are also varying degrees of improvement for medium and large targets. The comparative analysis shows that the improved YOLOv8n model performs better in detecting targets of different sizes, especially in the detection of small and medium targets. This proves that the proposed AMW-YOLOv8n is particularly beneficial for detecting road objects of various sizes.

Table 4. Average precision (AP) and average recall (AR) of YOLOv8n and AMW-YOLOv8n for large, medium, and small targets.

Models	AP Small/%	AP Medium/%	AP Large/%	AR Small/%	AR Medium/%	AR Large/%
YOLOv8n	35.6	63.3	74.6	43.0	60.7	68.9
AMW-YOLOv8n	45.6	66.4	78.6	55.7	67.7	71.3

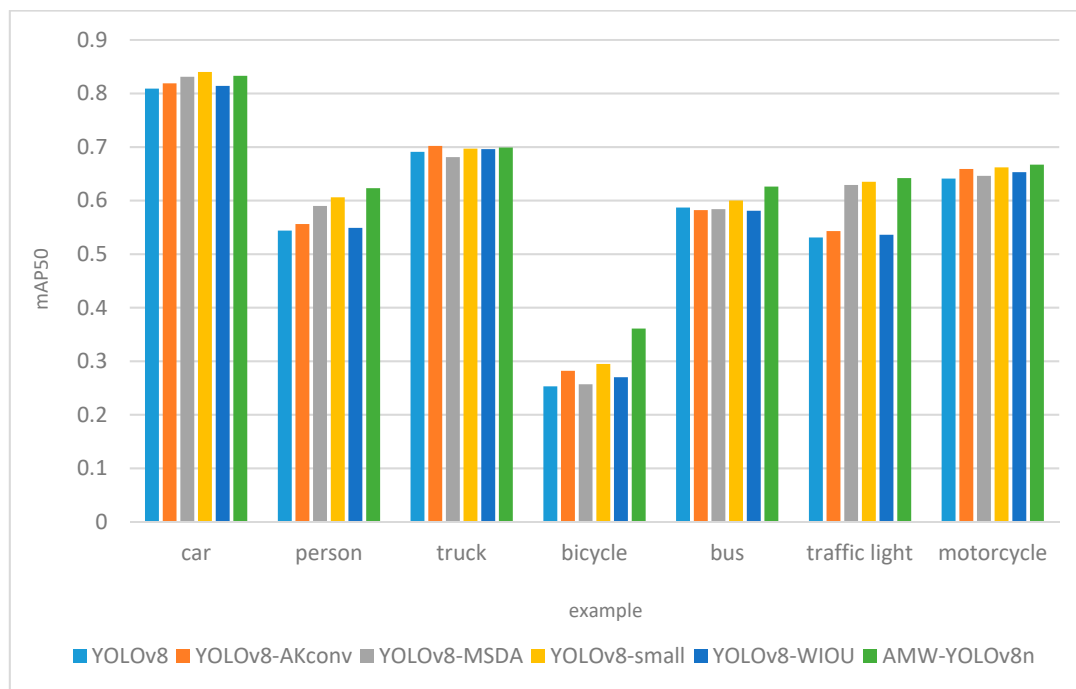


Figure 6. Comparative results of mean average precision (mAP) on different targets.

5. Conclusions

This study introduces the AMW-YOLOv8n algorithm for object detection in road scenes. Key contributions include enhancements to the YOLOv8 algorithm, such as AKconv, which dynamically adjusts the convolution kernel's shape and size to better adapt to features of varying scales, enhancing the network's capability to manage complex road targets. Furthermore, a Multi-Scale Dilated Attention mechanism is integrated to focus on key target features and enhance feature representation while maintaining computational efficiency. To address small and hard-to-recognize targets in complex traffic scenes, a layer for detecting small targets is incorporated, significantly boosting detection accuracy. Finally, a regression loss function with a Wise-IoU dynamic non-monotonic focusing mechanism is introduced to enhance training efficiency, improving the network's attention to standard quality anchor boxes, increasing prediction accuracy, and reducing missed and false detections.

Although the proposed algorithm achieves relatively high accuracy, there is still potential for improvement in precision and other aspects. Deploying the algorithm to edge devices with limited computing power also remains a significant challenge. Future work will focus on enhancing the detection accuracy of lightweight algorithms.

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